

FIGURE 5: Secondary Endpoint Design (Parameter Recovery & Predictive Equivalence)

A Parameter Recovery Contract at Anchor $\mathbf{x}_0 = (250, 0, 0, 0, 0)$ 1. Mass Scaling Exponent p_{mass} (156 Held-Out Cases)

- **Operator:** $p_{\text{mass}}(\hat{g}) = \frac{\partial \ln \hat{g}}{\partial \ln m} \Big|_{\mathbf{x}_0} = \frac{m}{\hat{g}} \frac{\partial \hat{g}}{\partial m} \Big|_{\mathbf{x}_0}$
- **Planted Truth:** $p_{\text{truth}} = 0.50$ (F01..F05, F08..F12, F17, F18) or $[0.45, 0.75]$ (F07)
- **Tolerance:** $|p_{\text{mass}}(\hat{g}) - p_{\text{truth}}| \leq 0.15$ ($\pm 30\%$ physical resolution)
- **Properties:** Scale-invariant to $A > 0$; algebraically invariant to $\sqrt{m/250}$ vs $\sqrt{m}$

2. Normalized Descriptor Coupling c_{desc} (84 Descriptor Cases)

- **Affine/Sat:** $c_{\text{desc}} = \frac{1}{\hat{g}} \frac{\partial \hat{g}}{\partial d} \Big|_{\mathbf{x}_0}$ (F08, F09, F11, F12, F17)
- **Interaction:** $c_{\text{desc}} = \frac{1}{\hat{g}} \frac{\partial^2 \hat{g}}{\partial d \partial d} \Big|_{\mathbf{x}_0}$ (F10) | **Exp:** $c_{\text{desc}} = \frac{3}{\hat{g}} \frac{\partial \hat{g}}{\partial d} \Big|_{\mathbf{x}_0}$ (F18)
- **Planted Truth:** $c_{\text{truth}} \in [0.25, 0.55]$ | **Tolerance:** $|c_{\text{desc}}(\hat{g}) - c_{\text{truth}}| \leq 0.10$
- **Reporting:** Joint /156, Mass Exponent /156, Descriptor Coupling /84

B Predictive Equivalence Design (12 Reference Covariate Frames)

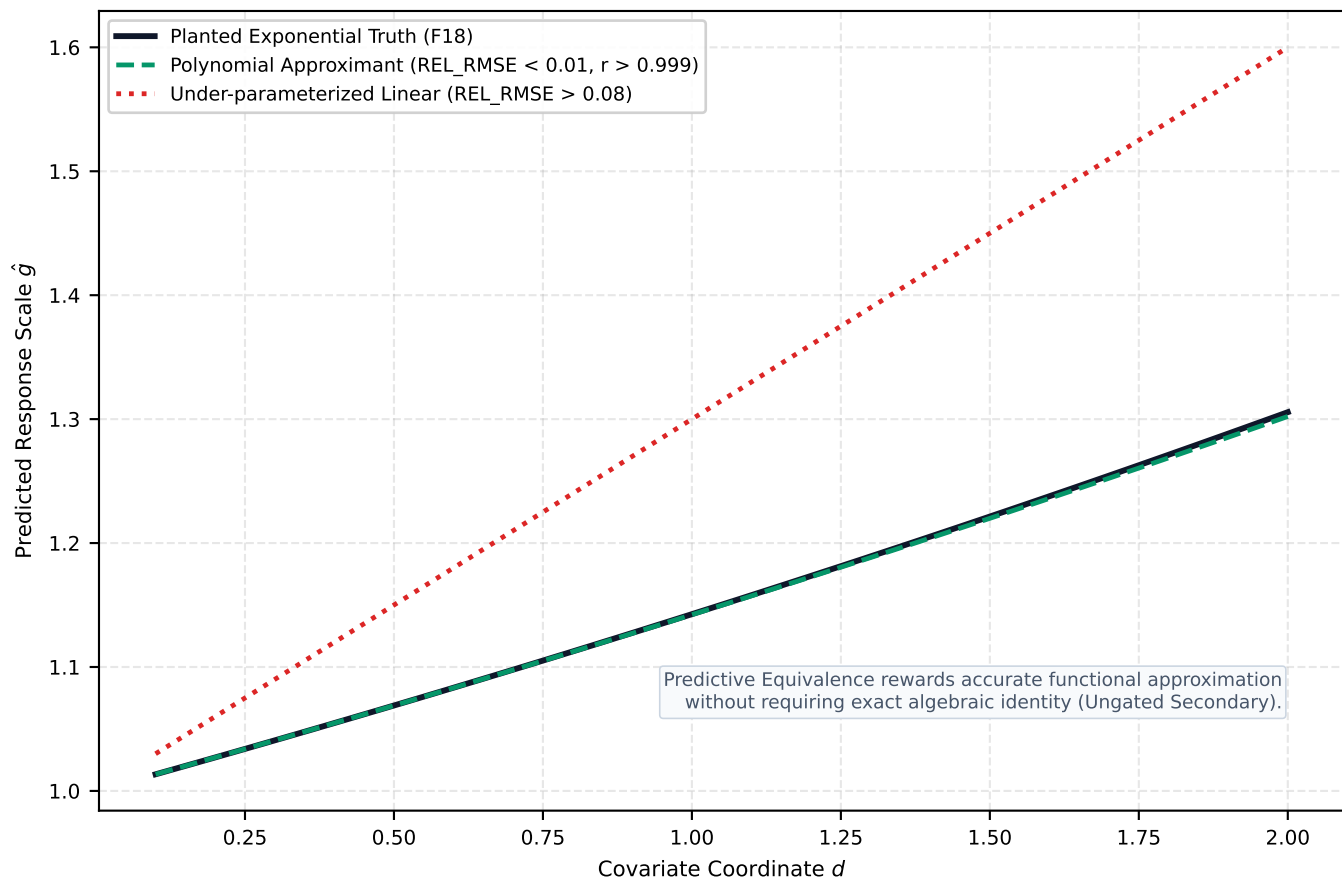
12 Generator-Derived Reference Frames (2,160 Rows)

- **Authentic Geometry:** 12 frames $\times$ 180 rows = 2,160 evaluation points
- **Scaffolds:** 30 scaffold groups/frame (6 compounds/scaffold)
- **Preserved Covariance:** Retains authentic $r \approx 0.98$ collinearity & finite-frame bounds
- **Canonical Digest:** SHA-256 **4fef2379ae33a10d089bd66794fdd21418b2b3...**

Scale-Invariant Out-of-Sample Metrics & Gates

1. **Valid Fraction:** valid fraction ≥ 0.995 ($|V| \geq 2150$ positive finite points)
2. **Scale Alignment:** $c^* = \frac{\sum y_i}{\sum y_i^2} > 0$ (Strictly positive multiplier)
3. **Relative RMSE:** $\text{REL_RMSE} = \frac{\sqrt{\frac{1}{N} \sum (c \cdot y_i - y_{\text{true},i})^2}}{\sqrt{\frac{1}{N} \sum y_i^2}} \leq 0.05$ (at most 5% error)
4. **Pearson Correlation:** $r \geq 0.990$ (Zero-variance rule: $r = 0.0 \rightarrow \text{FAIL}$)
5. **Denominator:** 144 Held-Out Cases (F01..F05, F08..F12, F17, F18)

C Predictive Fit vs Exact Algebra Distinction



D Observed Secondary Endpoint Outcomes

[PROSPECTIVE RESULT PANEL — DO NOT RENDER]

- Parameter Recovery Rate (Denominator: 156)
- Mass Exponent Recovery Decomposition (Denominator: 156)
- Descriptor Coupling Recovery Decomposition (Denominator: 84)
 - Predictive Equivalence Pass Rate (Denominator: 144)
 - Exact Algebra Recovery Rate (Denominator: 60)

All prospective outcomes remain sealed until one-shot execution.